

NAME

NETWORK PROTOCOLS LAB NO.

ID NO.

On the Header, put your **NAME**, **LAB Number** and **SMU ID**

Notes:

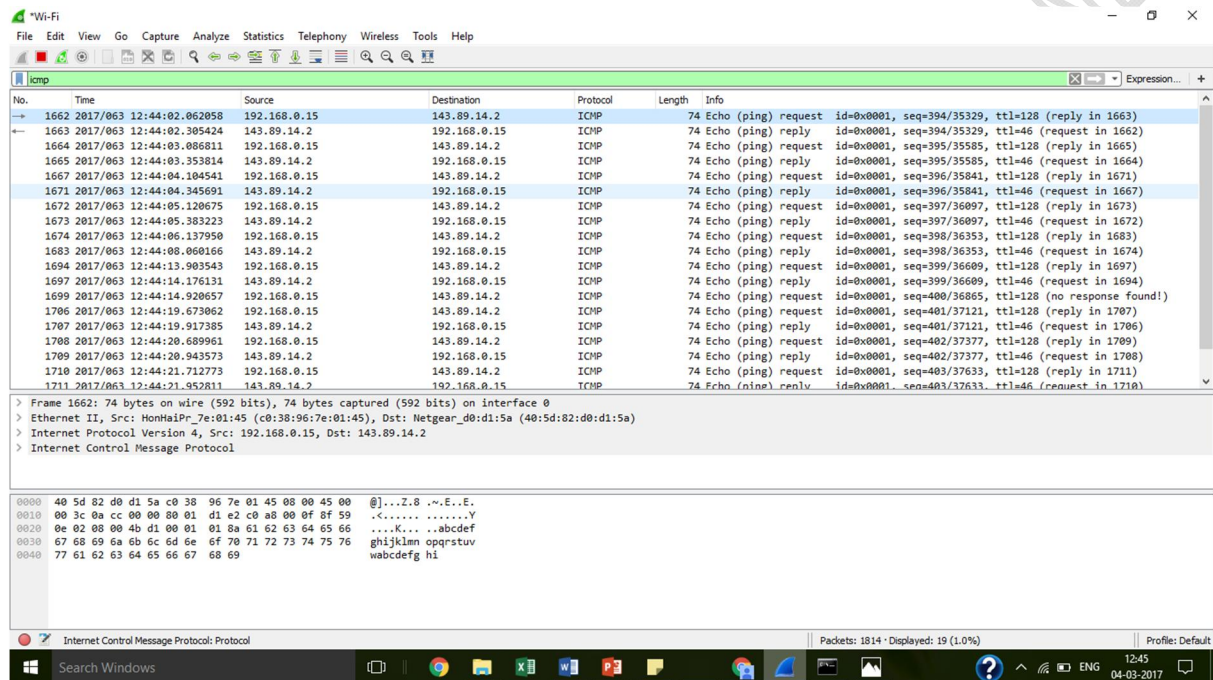
- The pattern is given to bring uniformity and should be followed for all Labs
- The assignment Questions should go in the way they are given, not to Skip any.
- Screenshots are mandatory with each answer
- Font colour: All Questions should be in **Bold Black** and Answers in **Black/Red**. While the lab report should be in Black only.
- For entire lab and lab report Font size is 12 and font style can be Arial/Calibri/TimesNewRoman
- Lab Report should be minimum of 2 pages and should be in your own words, not to be copied, plagiarized. (Give credits for figures, diagrams if you have taken from other source) You can include Block Diagrams, some figures to explain better or describe with some screenshots of the lab you performed. Lab report should come after the lab performed.
- The lab report should cover the concepts you have learned while performing the lab, do not repeat the steps you gone through, cover the purpose of the lab. You can write in paragraphs or bullet format.
- While submitting your assignment, the lab performed and the report both should be in the same file only. File format is DOC or PDF only. Don't submit in other formats.

Put **PAGE NUMBER** in the footer

WIRESHARK LAB NAME**1. What is the IP address of your host? What is the IP address of the destination host?**

Ans: The IP address of my host is 192.168.0.15

The IP address of my Destination host is 143.89.14.2

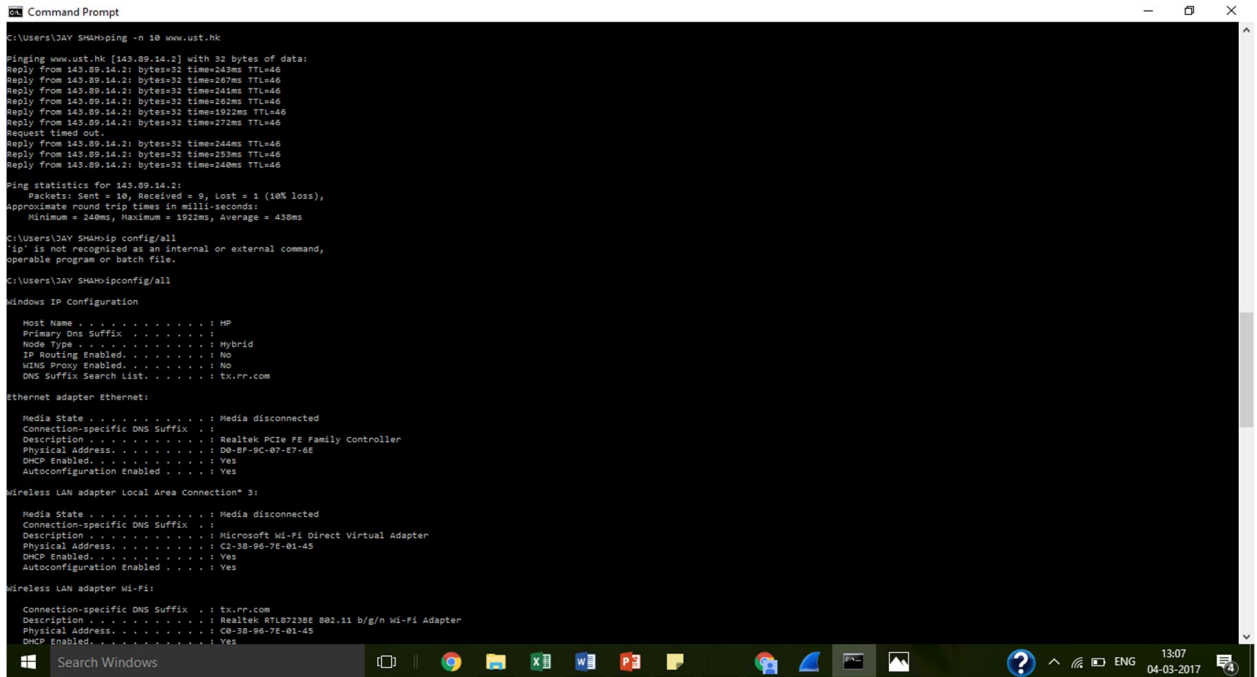


The screenshot shows a Wireshark capture of ICMP Echo (ping) traffic. The packet list on the left shows a series of requests and replies. The packet details pane on the right shows the structure of a selected packet (Frame 1662), including Ethernet II, Internet Protocol Version 4, and Internet Control Message Protocol (ICMP) fields. The packet bytes pane at the bottom shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1662	2017/06/3 12:44:02.062058	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=394/35329, ttl=128 (reply in 1663)
1663	2017/06/3 12:44:02.305424	143.89.14.2	192.168.0.15	ICMP	74	Echo (ping) reply id=0x0001, seq=394/35329, ttl=46 (request in 1662)
1664	2017/06/3 12:44:03.086811	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=395/35585, ttl=128 (reply in 1665)
1665	2017/06/3 12:44:03.353814	143.89.14.2	192.168.0.15	ICMP	74	Echo (ping) reply id=0x0001, seq=395/35585, ttl=46 (request in 1664)
1666	2017/06/3 12:44:04.104541	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=396/35841, ttl=128 (reply in 1671)
1671	2017/06/3 12:44:04.345691	143.89.14.2	192.168.0.15	ICMP	74	Echo (ping) reply id=0x0001, seq=396/35841, ttl=46 (request in 1667)
1672	2017/06/3 12:44:05.120675	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=397/36097, ttl=128 (reply in 1673)
1673	2017/06/3 12:44:05.383223	143.89.14.2	192.168.0.15	ICMP	74	Echo (ping) reply id=0x0001, seq=397/36097, ttl=46 (request in 1672)
1674	2017/06/3 12:44:06.137950	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=398/36353, ttl=128 (reply in 1683)
1683	2017/06/3 12:44:08.060166	143.89.14.2	192.168.0.15	ICMP	74	Echo (ping) reply id=0x0001, seq=398/36353, ttl=46 (request in 1674)
1694	2017/06/3 12:44:13.903543	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=399/36609, ttl=128 (reply in 1697)
1697	2017/06/3 12:44:14.176131	143.89.14.2	192.168.0.15	ICMP	74	Echo (ping) reply id=0x0001, seq=399/36609, ttl=46 (request in 1694)
1699	2017/06/3 12:44:14.920657	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=400/36865, ttl=128 (no response found!)
1706	2017/06/3 12:44:19.673062	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=401/37121, ttl=128 (reply in 1707)
1707	2017/06/3 12:44:19.917385	143.89.14.2	192.168.0.15	ICMP	74	Echo (ping) reply id=0x0001, seq=401/37121, ttl=46 (request in 1706)
1708	2017/06/3 12:44:20.689961	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=402/37377, ttl=128 (reply in 1709)
1709	2017/06/3 12:44:20.943573	143.89.14.2	192.168.0.15	ICMP	74	Echo (ping) reply id=0x0001, seq=402/37377, ttl=46 (request in 1708)
1710	2017/06/3 12:44:21.712773	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=403/37633, ttl=128 (reply in 1711)
1711	2017/06/3 12:44:21.952811	143.89.14.2	192.168.0.15	ICMP	74	Echo (ping) reply id=0x0001, seq=403/37633, ttl=46 (request in 1710)

Frame 1662: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 0
 Ethernet II, Src: HonHaiPr_7e:01:45 (c0:38:96:7e:01:45), Dst: Netgear_d0:d1:5a (40:5d:82:d0:d1:5a)
 Internet Protocol Version 4, Src: 192.168.0.15, Dst: 143.89.14.2
 Internet Control Message Protocol

0000 40 5d 82 d0 d1 5a c0 38 96 7e 01 45 08 00 45 00 @...2.8 ..E..
 0010 00 3c 0a cc 00 00 00 01 d1 e2 c0 a8 00 0f 8f 59 <.....Y
 0020 0e 02 08 00 4b d1 00 01 01 8a 61 62 63 64 65 66K...abcdef
 0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmn opqrstuv
 0040 77 61 62 63 64 65 66 67 68 69 wabcedfg hi



```
C:\Users\JAY>SHMOping -n 10 www.ust.hk

Pinging www.ust.hk [143.89.14.2] with 32 bytes of data:
Reply from 143.89.14.2: bytes=32 time=243ms TTL=46
Reply from 143.89.14.2: bytes=32 time=267ms TTL=46
Reply from 143.89.14.2: bytes=32 time=241ms TTL=46
Reply from 143.89.14.2: bytes=32 time=262ms TTL=46
Reply from 143.89.14.2: bytes=32 time=192ms TTL=46
Reply from 143.89.14.2: bytes=32 time=272ms TTL=46
Request timed out.
Reply from 143.89.14.2: bytes=32 time=244ms TTL=46
Reply from 143.89.14.2: bytes=32 time=253ms TTL=46
Reply from 143.89.14.2: bytes=32 time=248ms TTL=46

Ping statistics for 143.89.14.2:
    Packets: Sent = 10, Received = 9, Lost = 1 (10% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 248ms, Maximum = 292ms, Average = 438ms

C:\Users\JAY>SHMOipconfig/all
'ip' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\JAY>SHMOipconfig/all

Windows IP Configuration

Host Name . . . . . : HP
Primary Dns Suffix . . . . . : 
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : tx.rp.com

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Realtek PCIe FE Family Controller
Physical Address. . . . . : 08-00-30-07-07-04
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 3:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 08-00-30-07-07-04
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : tx.rp.com
Description . . . . . : Realtek RTL8723BE 802.11 b/g/n Wi-Fi Adapter
Physical Address. . . . . : 08-00-30-07-07-04
DHCP Enabled. . . . . : Yes
```

2. Why is it that an ICMP packet does not have source and destination port numbers?

Ans. ICMP gives information between routers and hosts i.e. in the network layer. It does not communicate between the application layer process. No port numbers are needed to direct the ICMP message to application layer process as the network software itself interprets all ICMP messages. Each ICMP packet has a type and code which identifies the specific message being received.

3. Examine one of the ping request packets sent by your host. What are the ICMP type and code numbers? What other fields does this ICMP packet have? How many bytes are the checksum, sequence number and identifier fields?

Ans.

The ICMP Type is 8 and Code is 0.

The other fields that this ICMP packet has are checksum, identifier, sequence number and data fields.

The checksum, sequence number and identifier fields are of 2 bytes each.

Wireshark packet capture showing ICMP Echo (ping) request and reply. The packet list shows a request from 192.168.0.15 to 143.89.14.2 and a reply from 143.89.14.2 to 192.168.0.15. The packet details pane shows the ICMP Echo (ping) request with type 8, code 0, and sequence number 395. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1662	2017/06/3 12:44:02.062058	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=394/35329, ttl=128 (reply in 1663)
1664	2017/06/3 12:44:03.086811	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=395/35585, ttl=128 (reply in 1665)
1667	2017/06/3 12:44:04.104541	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=396/35841, ttl=128 (reply in 1671)
1672	2017/06/3 12:44:05.120675	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=397/36097, ttl=128 (reply in 1673)
1674	2017/06/3 12:44:06.137950	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=398/36353, ttl=128 (reply in 1683)
1694	2017/06/3 12:44:13.903543	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=399/36609, ttl=128 (reply in 1697)
1699	2017/06/3 12:44:14.920657	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=400/36865, ttl=128 (no response found!)
1706	2017/06/3 12:44:19.673062	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=401/37121, ttl=128 (reply in 1707)
1708	2017/06/3 12:44:20.689961	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=402/37377, ttl=128 (reply in 1709)
1710	2017/06/3 12:44:21.712773	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=403/37633, ttl=128 (reply in 1711)
1663	2017/06/3 12:44:02.305424	143.89.14.2	192.168.0.15	ICMP	74	Echo (ping) reply id=0x0001, seq=394/35329, ttl=46 (request in 1662)
1665	2017/06/3 12:44:03.353814	143.89.14.2	192.168.0.15	ICMP	74	Echo (ping) reply id=0x0001, seq=395/35585, ttl=46 (request in 1664)
1671	2017/06/3 12:44:04.345691	143.89.14.2	192.168.0.15	ICMP	74	Echo (ping) reply id=0x0001, seq=396/35841, ttl=46 (request in 1667)

Frame 1664: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 0
 Ethernet II, Src: MonksiaPr_7e:01:45 (c0:38:96:7e:01:45), Dst: Netgear_d0:d1:5a (40:5d:82:d0:d1:5a)
 Internet Protocol Version 4, Src: 192.168.0.15, Dst: 143.89.14.2
 Internet Control Message Protocol
 Type: 8 (Echo (ping) request)
 Code: 0
 Checksum: 0x4bd0 [correct]
 [Checksum Status: Good]
 Identifier (BE): 1 (0x0001)
 Identifier (LE): 256 (0x0100)
 Sequence number (BE): 395 (0x018b)
 Sequence number (LE): 35585 (0x8b01)
 [Response frame: 1665]
 Data (32 bytes)

0000 40 5d 82 d0 d1 5a c0 38 96 7e 01 45 00 00 45 00 @...Z.0 ~.E..E.
 0010 00 3c 0a cd 00 00 00 01 d1 e1 c0 a8 00 0f 8f 59 <.....Y
 0020 0e 02 00 00 4b d0 00 01 01 8b 61 62 63 64 65 66 ..K....abcdef
 0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmn opqrstuv
 0040 77 61 62 63 64 65 66 67 68 69 wabcedfg hi

4. Examine the corresponding ping reply packet. What are the ICMP type and code numbers? What other fields does this ICMP packet have? How many bytes are the checksum, sequence number and identifier fields?

Ans. The corresponding ping reply packet is examined and I came to know that the ICMP Type and Code both is Zero 0. The other packets this packet has been Checksum, Identifier, Data and Sequence number. The checksum, sequence number and identifier fields all are of 2 bytes each.

The screenshot shows a Wireshark packet capture of ICMP Echo (ping) traffic. The packet list shows multiple requests and replies. The selected packet is a reply from 192.168.0.15 to 143.89.14.2.

No.	Time	Source	Destination	Protocol	Length	Info
1662	2017/06/3 12:44:02.062058	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=394/35329, ttl=128 (reply in 1663)
1664	2017/06/3 12:44:03.086811	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=395/35585, ttl=128 (reply in 1665)
1667	2017/06/3 12:44:04.104541	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=396/35841, ttl=128 (reply in 1671)
1672	2017/06/3 12:44:05.120675	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=397/36097, ttl=128 (reply in 1673)
1674	2017/06/3 12:44:06.137950	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=398/36353, ttl=128 (reply in 1683)
1694	2017/06/3 12:44:13.903543	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=399/36609, ttl=128 (reply in 1697)
1699	2017/06/3 12:44:14.920657	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=400/36865, ttl=128 (no response found!)
1706	2017/06/3 12:44:19.673062	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=401/37121, ttl=128 (reply in 1707)
1708	2017/06/3 12:44:20.689961	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=402/37377, ttl=128 (reply in 1709)
1710	2017/06/3 12:44:21.712773	192.168.0.15	143.89.14.2	ICMP	74	Echo (ping) request id=0x0001, seq=403/37633, ttl=128 (reply in 1711)
1663	2017/06/3 12:44:02.305424	143.89.14.2	192.168.0.15	ICMP	74	Echo (ping) reply id=0x0001, seq=394/35329, ttl=46 (request in 1662)
1665	2017/06/3 12:44:03.353814	143.89.14.2	192.168.0.15	ICMP	74	Echo (ping) reply id=0x0001, seq=395/35585, ttl=46 (request in 1664)
1671	2017/06/3 12:44:04.345691	143.89.14.2	192.168.0.15	ICMP	74	Echo (ping) reply id=0x0001, seq=396/35841, ttl=46 (request in 1667)

Frame 1665: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 0
 Ethernet II, Src: Netgear_d0:d1:5a (40:5d:82:d0:d1:5a), Dst: MonHaiPr_7e:01:45 (c0:38:96:7e:01:45)
 Internet Protocol Version 4, Src: 143.89.14.2, Dst: 192.168.0.15
 Internet Control Message Protocol
 Type: 0 (Echo (ping) reply)
 Code: 0
 Checksum: 0x53d0 [correct]
 [Checksum Status: Good]
 Identifier (BE): 1 (0x0001)
 Identifier (LE): 256 (0x0100)
 Sequence number (BE): 395 (0x018b)
 Sequence number (LE): 35585 (0x8b01)
 [Request frame: 1664]
 [Response time: 267.003 ms]
 Data (32 bytes)

0000 c0 38 96 7e 01 45 40 5d 82 d0 d1 5a 00 00 45 00 .8~.E[] ...Z..E.
 0010 00 3c ef 12 00 00 2e 01 3f 9c 8f 59 0e 02 c0 a8 .<.....?.Y.....
 0020 00 0f 00 00 53 d0 00 01 01 8b 61 62 63 64 65 66 ..S....abcdef
 0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmn opqrstuv
 0040 77 61 62 63 64 65 66 67 68 69 wabcedfg hi

5. What is the IP address of your host? What is the IP address of the target destination host?

Ans. The IP address of my host is 192.168.0.15

The IP address of my Target Destination host is 128.93.162.84

```
Command Prompt
DHCPv6 Client GUID: . . . . . : 00-01-00-01-1C-08-92-25-08-8F-9C-07-E7-6E
NetBIOS over Tcpip. . . . . : Disabled

C:\Users\JAY SHAM>www.inria.fr
'www.inria.fr' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\JAY SHAM>tracert www.inria.fr
Tracing route to esp3.inria.fr [128.93.162.84]
over a maximum of 30 hops:
  0  1 ms  1 ms  1 ms  192.168.0.1
  1  10 ms  25 ms  11 ms  142.254.160.133
  2  15 ms  27 ms  *  tpea-0-0-dl1tx5102h.texas.rr.com [24.28.89.105]
  3  118 ms  26 ms  13 ms  agg34.dl1btx1b02r.texas.rr.com [24.175.50.1]
  4  17 ms  22 ms  21 ms  agg31.netb1308r.texas.rr.com [24.175.49.8]
  5  16 ms  21 ms  29 ms  at-1-0-p0.at190.tbone.rr.com [66.109.1.216]
  6  77 ms  77 ms  70 ms  bu-ether12.dl1tx9761w-bcr00.tbone.rr.com [66.109.6.39]
  7  65 ms  72 ms  70 ms  bu-ether23.dl1tx9761w-bcr00.tbone.rr.com [66.109.9.41]
  8  91 ms  78 ms  *  bu-ether11.nycmy837aw-bcr00.tbone.rr.com [66.109.6.24]
  9  167 ms  75 ms  71 ms  0.sai.pri.nyc20.tbone.rr.com [107.14.17.218]
 10  69 ms  67 ms  66 ms  66.109.27.26
 11  *  *  *  *
 12  *  *  *  *
 13  143 ms  138 ms  139 ms  et-10-3-0-xcri.ptl.cw.net [195.2.24.242]
 14  156 ms  148 ms  153 ms  as3-xcri.ppp.cw.net [195.2.10.88]
 15  142 ms  148 ms  *  giprenater-gu.par.cw.net [195.10.54.66]
 16  258 ms  138 ms  140 ms  te2-1-paris1-rtr-021.noc.renater.fr [193.51.177.27]
 17  138 ms  144 ms  138 ms  te1-1-inria-rtr-021.noc.renater.fr [193.51.177.107]
 18  143 ms  144 ms  150 ms  inria-rocquencourt-te1-4-inria-rtr-021.noc.renater.fr [193.51.184.177]
 19  153 ms  143 ms  147 ms  unit240-reth1-vfw-ext-dci.inria.fr [192.93.122.19]
 20  174 ms  142 ms  140 ms  esp3.inria.fr [128.93.162.84]

Trace complete.

C:\Users\JAY SHAM>
```

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter: <Ctrl>+> Expression...

Time	Source	Destination	Protocol	Length	Info
22 2017/06/3 13:57:51.975372	192.168.0.15	128.93.162.84	ICMP	106	Echo (ping) request id=0x0001, seq=464/53249, ttl=1 (no response found!)
23 2017/06/3 13:57:51.975596	192.168.0.15	128.93.162.84	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
24 2017/06/3 13:57:51.977356	192.168.0.15	128.93.162.84	ICMP	106	Echo (ping) request id=0x0001, seq=465/53585, ttl=1 (no response found!)
25 2017/06/3 13:57:51.978255	192.168.0.15	128.93.162.84	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
26 2017/06/3 13:57:51.978923	192.168.0.15	128.93.162.84	ICMP	106	Echo (ping) request id=0x0001, seq=466/53761, ttl=1 (no response found!)
27 2017/06/3 13:57:51.979814	192.168.0.15	128.93.162.84	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)

< Frame 22: 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on interface 0

> Ethernet II, Src: HonHaiPr_7e:01:45 (c0:38:96:7e:01:45), Dst: Netgear_d0:d1:5a (40:5d:82:d0:d1:5a)

> Internet Protocol Version 4, Src: 192.168.0.15, Dst: 128.93.162.84

> Internet Control Message Protocol

Type: 8 (Echo (ping) request)

Code: 0

Checksum: 0xf62e [correct]

[Checksum Status: Good]

Identifier (BE): 1 (0x0001)

Identifier (LE): 256 (0x0100)

Sequence number (BE): 464 (0x01d0)

Sequence number (LE): 53249 (0xd001)

> [No response seen]

> Data (64 bytes)

0000 40 5d 82 d0 d1 5a c0 38 96 7e 01 45 00 00 45 00 @...Z.B...E..

0010 00 5c 4e 81 00 00 01 01 87 b7 c0 a8 0f 80 5d .\N.....

0020 a2 54 00 00 f6 2e 00 01 01 d0 00 00 00 00 00 .J.....

0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00

0040 00 00 00 00 00 00 00 00 00 00 00 00 00 00

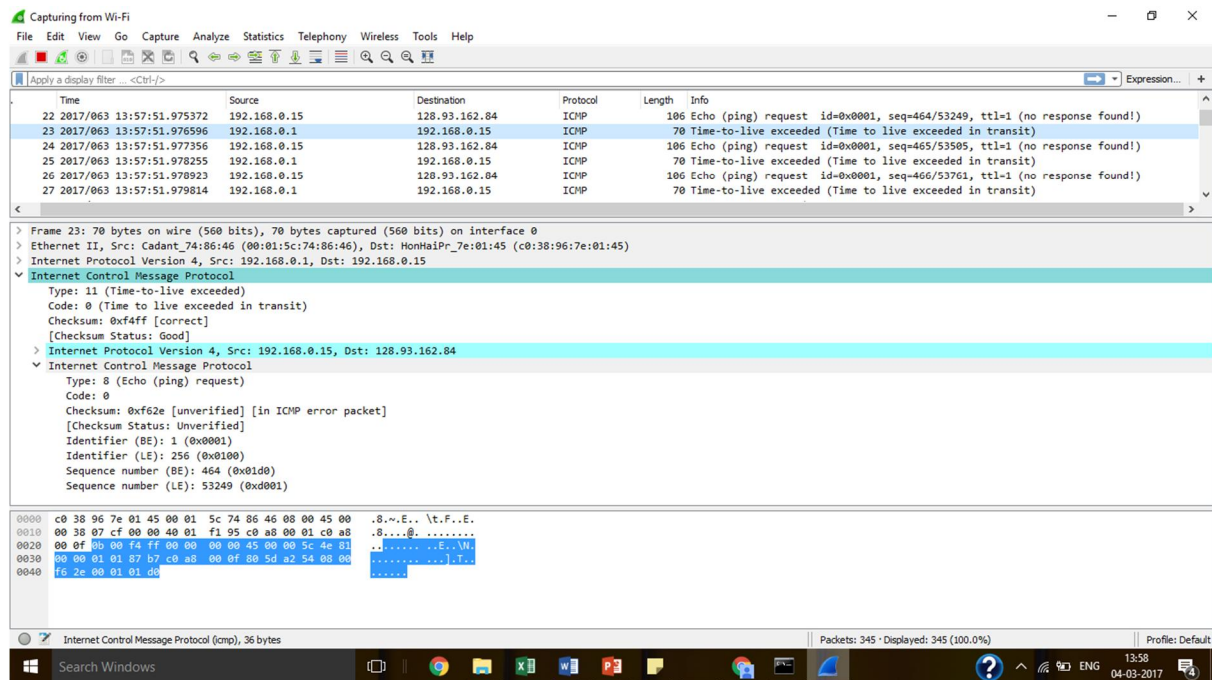
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00

0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00

Internet Control Message Protocol (icmp), 72 bytes

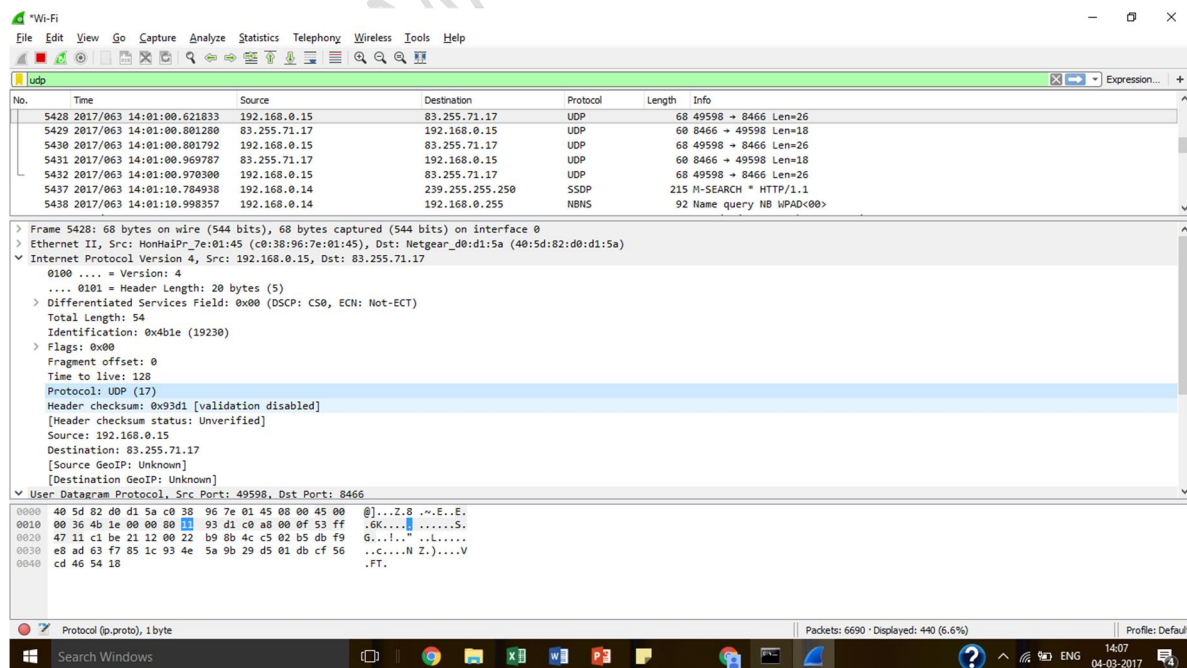
Packets: 587 - Displayed: 587 (100.0%)

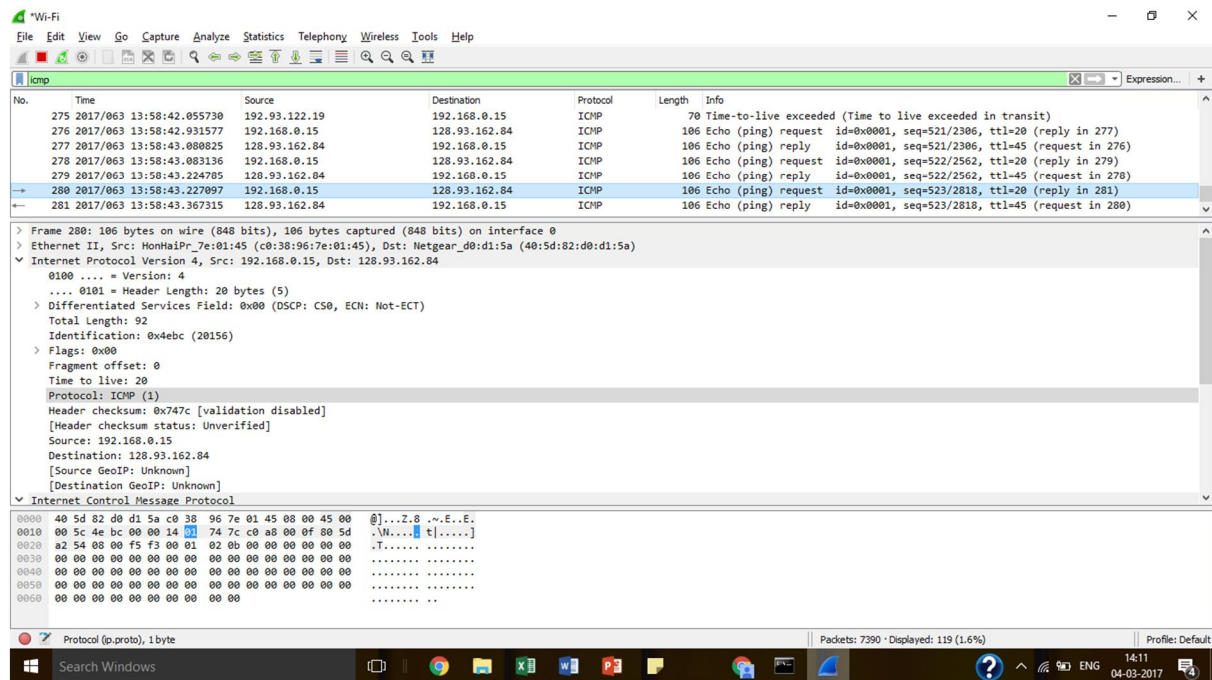
Profile: Default



6. If ICMP sent UDP packets instead (as in Unix/Linux), would the IP protocol number still be 01 for the probe packets? If not, what would it be?

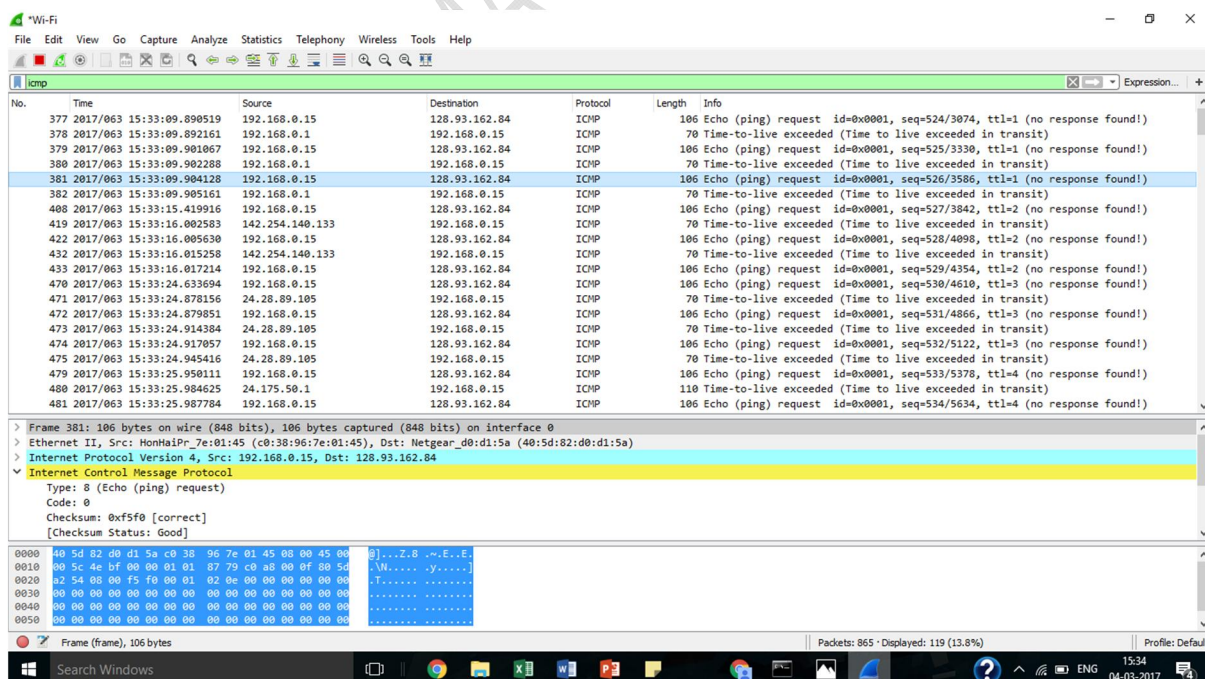
Ans. If ICMP sent UDP packets are sent then the IP protocol number for UDP would be 17 and it won't be 01 for the probe packets.





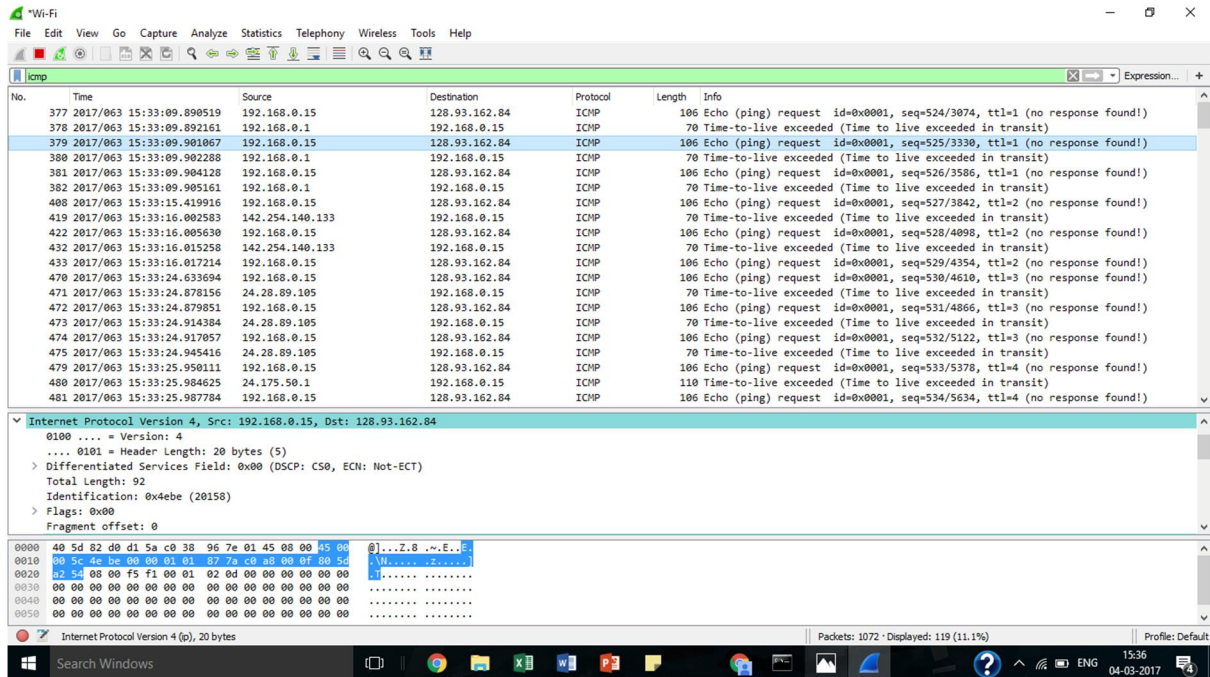
7. Examine the ICMP echo packet in your screenshot. Is this different from the ICMP ping query packets in the first half of this lab? If yes, how so?

Ans. The ICMP ping query packets in the first half of this lab is the same as the ICMP echo packet



8. Examine the ICMP error packet in your screenshot. It has more fields than the ICMP echo packet. What is included in those fields?

Ans. It includes the IP header which contains all ipv4 fields as well as the first 8 bytes of the original ICMP packet that the error is for which contains information about the packet like the type, code, identifier, sequence number and checksum fields.



9. Examine the last three ICMP packets received by the source host. How are these packets different from the ICMP error packets? Why are they different?

Ans. The last three ICMP packets received by the source host have are type 0 unlike the error packets which are type 8. They are different because the packets have reached the destination host before TTL expired.

NAME

NETWORK PROTOCOLS LAB NO.

ID NO.

Wi-Fi

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icmp

No.	Time	Source	Destination	Protocol	Length	Info
628	2017/06/3 15:33:51.477818	192.168.0.15	128.93.162.84	ICMP	106	Echo (ping) request id=0x0001, seq=573/15618, ttl=17 (no response found!)
629	2017/06/3 15:33:51.618858	193.51.177.107	192.168.0.15	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
630	2017/06/3 15:33:51.621282	192.168.0.15	128.93.162.84	ICMP	106	Echo (ping) request id=0x0001, seq=574/15874, ttl=17 (no response found!)
631	2017/06/3 15:33:51.761594	193.51.177.107	192.168.0.15	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
636	2017/06/3 15:33:52.791085	192.168.0.15	128.93.162.84	ICMP	106	Echo (ping) request id=0x0001, seq=575/16130, ttl=18 (no response found!)
637	2017/06/3 15:33:52.942521	193.51.184.177	192.168.0.15	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
638	2017/06/3 15:33:52.944674	192.168.0.15	128.93.162.84	ICMP	106	Echo (ping) request id=0x0001, seq=576/16386, ttl=18 (no response found!)
639	2017/06/3 15:33:53.086700	193.51.184.177	192.168.0.15	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
640	2017/06/3 15:33:53.088944	192.168.0.15	128.93.162.84	ICMP	106	Echo (ping) request id=0x0001, seq=577/16642, ttl=18 (no response found!)
641	2017/06/3 15:33:53.231889	193.51.184.177	192.168.0.15	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
646	2017/06/3 15:33:54.378370	192.168.0.15	128.93.162.84	ICMP	106	Echo (ping) request id=0x0001, seq=578/16898, ttl=19 (no response found!)
647	2017/06/3 15:33:58.166375	192.168.0.15	128.93.162.84	ICMP	106	Echo (ping) request id=0x0001, seq=579/17154, ttl=19 (no response found!)
659	2017/06/3 15:34:02.157625	192.168.0.15	128.93.162.84	ICMP	106	Echo (ping) request id=0x0001, seq=580/17410, ttl=19 (no response found!)
660	2017/06/3 15:34:02.402731	192.93.122.19	192.168.0.15	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
665	2017/06/3 15:34:03.457986	192.168.0.15	128.93.162.84	ICMP	106	Echo (ping) request id=0x0001, seq=581/17666, ttl=20 (reply in 666)
666	2017/06/3 15:34:03.604942	128.93.162.84	192.168.0.15	ICMP	106	Echo (ping) reply id=0x0001, seq=581/17666, ttl=45 (request in 665)
667	2017/06/3 15:34:03.606480	192.168.0.15	128.93.162.84	ICMP	106	Echo (ping) request id=0x0001, seq=582/17922, ttl=20 (reply in 668)
668	2017/06/3 15:34:03.745586	128.93.162.84	192.168.0.15	ICMP	106	Echo (ping) reply id=0x0001, seq=582/17922, ttl=45 (request in 667)
669	2017/06/3 15:34:03.747750	192.168.0.15	128.93.162.84	ICMP	106	Echo (ping) request id=0x0001, seq=583/18178, ttl=20 (reply in 670)
670	2017/06/3 15:34:03.898250	128.93.162.84	192.168.0.15	ICMP	106	Echo (ping) reply id=0x0001, seq=583/18178, ttl=45 (request in 669)

Internet Protocol Version 4, Src: 128.93.162.84, Dst: 192.168.0.15

Internet Control Message Protocol

Type: 0 (Echo (ping) reply)

Code: 0

Checksum: 0xfdb7 [correct]

[Checksum Status: Good]

Identifier (BE): 1 (0x0001)

Identifier (LE): 256 (0x0100)

0000 c0 38 96 7e 01 45 40 5d 82 d0 d1 5a 08 00 45 00 .8~.f0]...Z...
0010 00 5c 58 d0 00 00 2d 01 51 68 80 5d a2 54 c0 a8 .X.....0h].T...
0020 00 0f 00 00 fd b7 00 01 02 47 00 00 00 00 00 00 .G.....
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0040 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Internet Protocol Version 4 (ip), 20 bytes

Packets: 1371 · Displayed: 119 (8.7%)

Profile: Default

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icmp

No.	Time	Source	Destination	Protocol	Length	Info
628	2017/06/3 15:33:51.477818	192.168.0.15	128.93.162.84	ICMP	106	Echo (ping) request id=0x0001, seq=573/15618, ttl=17 (no response found!)
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666	2017/06/3 15:34:03.604942	128.93.162.84	192.168.0.15	ICMP	106	Echo (ping) reply id=0x0001, seq=581/17666, ttl=45 (request in 665)
667	2017/06/3 15:34:03.606480	192.168.0.15	128.93.162.84	ICMP	106	Echo (ping) request id=0x0001, seq=582/17922, ttl=20 (reply in 668)
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670	2017/06/3 15:34:03.898250	128.93.162.84	192.168.0.15	ICMP	106	Echo (ping) reply id=0x0001, seq=583/18178, ttl=45 (request in 669)

Internet Protocol Version 4, Src: 192.168.0.15, Dst: 128.93.162.84

Internet Control Message Protocol

Type: 8 (Echo (ping) request)

Code: 0

Checksum: 0xf5b7 [correct]

[Checksum Status: Good]

Identifier (BE): 1 (0x0001)

Identifier (LE): 256 (0x0100)

0000 40 5d 82 d0 d1 5a c0 38 96 7e 01 45 08 00 45 00 @]...Z.8~.E...
0010 00 5c 4e f8 00 00 14 01 74 40 c0 a8 0f 80 5d .W.....t@.....
0020 a2 54 00 00 f5 b7 00 01 02 47 00 00 00 00 00 00 .G.....
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0040 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Internet Protocol Version 4 (ip), 20 bytes

Packets: 1380 · Displayed: 119 (8.6%)

Profile: Default

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10. Within the tracert measurements, is there a link whose delay is significantly longer than others? Refer to the screenshot in Figure 4, is there a link whose delay is significantly longer than others? On the basis of the router names, can you guess the location of the two routers on the end of this link?

Ans. Total is 30 Hops

From the observation of trace route, we can say that the delay for the hop 12 is longer than hop 11. The router at hop 11 is located at USA and the router at hop 12 is located in Europe

```

C:\Users\JAY>tracert www.inria.fr
Tracing route to esp3.inria.fr [128.93.162.86]
over a maximum of 30 hops:
 0  0 ms  0 ms  0 ms  192.168.0.1
 1  10 ms  25 ms  11 ms  142.254.140.133
 2  15 ms  27 ms  *   tge0-0-0-dl1xt3102h.texas.rr.com [24.28.89.105]
 3  110 ms  20 ms  18 ms  agg24-dl1bt100r.texas.rr.com [24.179.56.1]
 4  17 ms  21 ms  21 ms  agg11-hstxt1100r.texas.rr.com [24.179.49.8]
 5  16 ms  21 ms  29 ms  ae-1-0-p0-at100.tbone.rr.com [66.109.1.218]
 6  77 ms  77 ms  78 ms  bu-ether12-dl1tx0761w-bcr00.tbone.rr.com [66.109.6.39]
 7  65 ms  72 ms  70 ms  bu-ether13-dl1tx0761w-bcr00.tbone.rr.com [66.109.9.41]
 8  91 ms  78 ms  *   bu-ether11-nycmy037aw-bcr00.tbone.rr.com [66.109.6.24]
 9  187 ms  75 ms  71 ms  0-ae1-pr1.nyc28.tbone.rr.com [107.14.17.218]
10  69 ms  67 ms  66 ms  66.109.7.16
11  *      160 ms  148 ms  aeb-xcr1.nyh.cw.net [195.2.25.70]
12  143 ms  138 ms  139 ms  et-10-3-0-xcr1.pil.cw.net [195.2.24.242]
13  156 ms  140 ms  150 ms  aeb-xcr1.prp.cw.net [195.2.10.89]
14  142 ms  140 ms  *   giprenater-gw.par.cw.net [195.10.54.66]
15  258 ms  138 ms  140 ms  te1-1-par11-rtr-021.noc.renater.fr [193.51.177.27]
16  138 ms  146 ms  130 ms  te1-1-inria-rtr-021.noc.renater.fr [193.51.177.107]
17  163 ms  144 ms  150 ms  inria-rocquencourt-te1-4-inria-rtr-021.noc.renater.fr [193.51.184.177]
18  153 ms  143 ms  147 ms  uni12ab-reth1-vf-went-dc1.inria.fr [192.93.122.19]
19  174 ms  142 ms  140 ms  esp3.inria.fr [128.93.162.86]

Trace complete.
C:\Users\JAY>

```

LAB REPORT

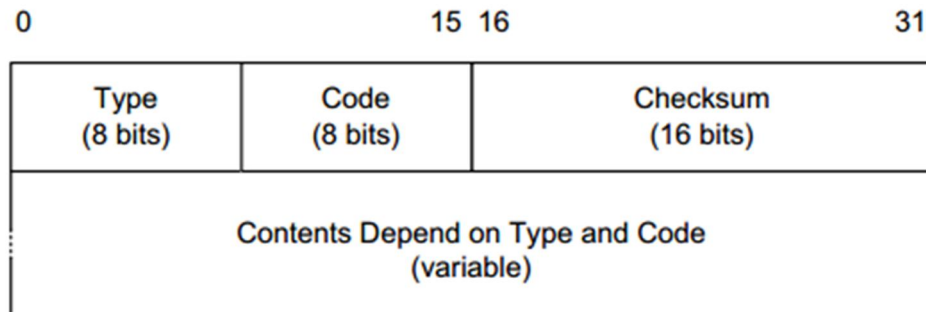
ICMP is also called as Internet Control Message Protocol. It is an error reporting protocol for network devices such as router which helps the device to notify the source that there is error in transmitting the information packet. ICMP messages can be sent, received or processed in an IP network. ICMP is used to trouble shoot internet connection. The messages are used by the routers or other intermediate devices to communicate with the source. Both IPv4 and IPv6 use the same version of ICMP protocol. Before transmitting, IP packet is encapsulated with ICMP data.

Query Mechanism is not present in IP protocol related to host and management. So ICMP was introduced to resolve problems related to network connectivity such that network administrator requires information from another router or host. At the source end host needs to know if a router or the receiving host is alive.

I came to know the following properties while performing the lab

- ICMP messages are generated by the Ping program
- Ping is used to check the devices such as router are alive in the network in the lab which study the ping programming
- Ping operates by sending ICMP Echo Requests packets to the target host and waiting for ICMP Echo Request
- The assigned protocol number is 1 for ICMP. ICMP differs from the transport protocols i.e UDP and TCP. TCP and UDP are not typically used to exchange data or can communicate between the systems.
- The IP port numbers for TCP and UDP is 6 and 17.
- Error reporting messages and query messages are the two types of ICMP messages. When an IP packets gets, problems related to routers or host is reported by error reporting message. Specific information of router and host is obtained using query messages.
- ICMP is a protocol for sending various messages to report network conditions. The echo request is one of many messages. Ping can be filtered but ICMP messages types are required for operation of various protocols.
- ICMP is implemented as a part of the IP layer so ICMP is a layer 3 protocol
- Checksum Calculation is a method to solve error detection. In checksum, all field are set to zero and all sums are of 16bits words. Checksum filed is stored and sum of all bits are complemented
- To determine the RTT, we need two machines that can use the time stamp request and time stamp reply message.
- The echo request and echo reply are used to diagnose the problems and the host and routers are determined by using echo request and reply.
- A destination unreachable message indicates that the datagram is not delivered. These messages is sent by the router or host when datagram is discarded. Reason for discarding the datagram is specified by code field.
- Some of the codes are: code 0 is due to hardware failure network, Code 1 is due to hardware failure host, Code 2 is when protocol is unreachable and Code 3 is the port is unreachable

- The Tracert in window can be used to trace the packets travelling from host device to destination device
- The formats and the contents of an ICMP message was cleared while studying ICMP



ICMP HEADER

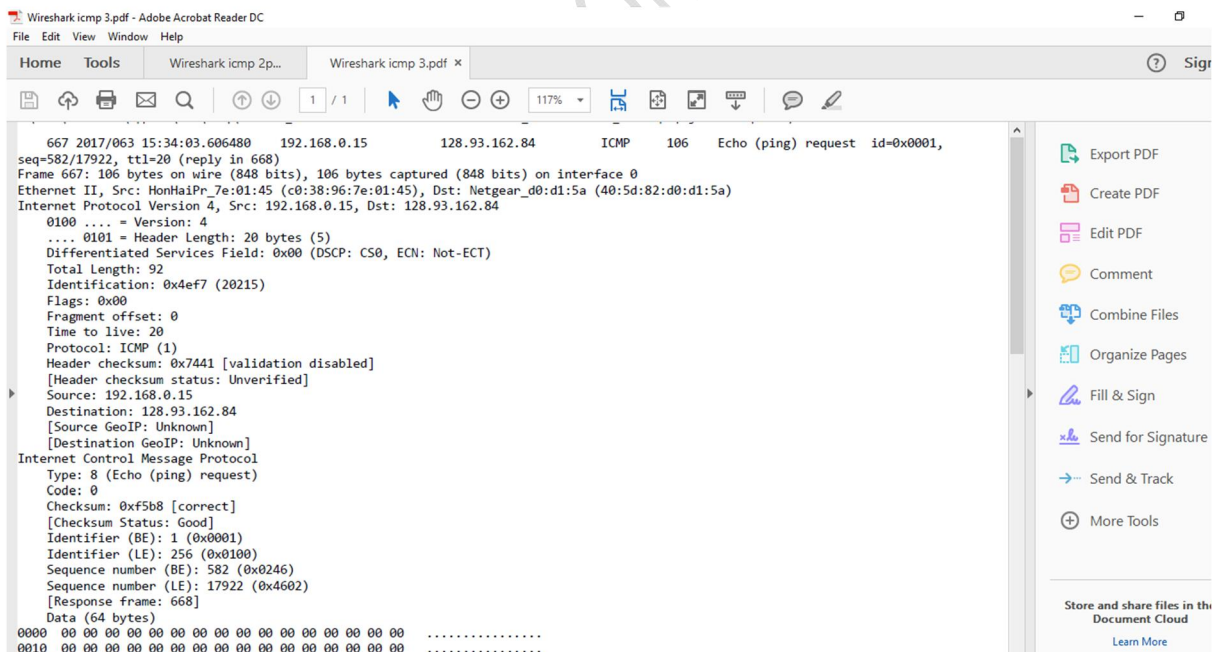
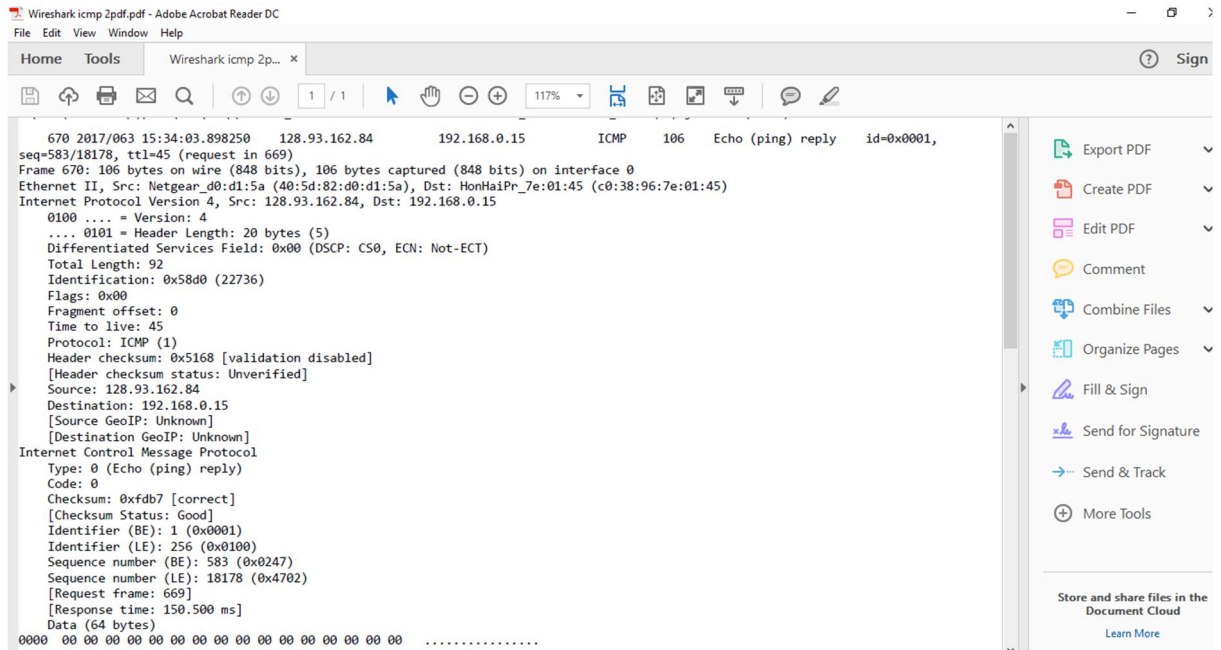
- Type field has 42 different value are reserved, which identify the message. 8 bits are for regular use.
- Code field to further specify the meaning of the message.
- The Checksum field covers the entire message.

Checking ICMP response that the router sends to host PC to inform the errors and queries.

The following are the error messages which I observed when I performed the lab.

- TTL- Time-to-live – Exceed – The value of the IP Packet reaches zero. A situation is arrived when a loop is created when there is error in the routing table, moving from one router to another and each datagram contains a field called TTL.
- Destination Unreached: The frame or piece was discarded or dropped because it could not reach the destination
- Redirect Message: If the datagram reaches to a router where it was not intended to go then the router checks the routing table and then redirects the datagram packet to the correct router
- Redirection: The code field for the redirection message narrows down the redirection.
- The codes are: Code 0: Redirection for a network specific route, Code 1: Redirection for host Specific route, Code 2 is based on the type of service reduction on network specific route. Code 3 is on type of service reduction based on specific route.
- Parameter Problem Messages- If there is any issue related to Checksum, Type, Code it reports to the source host. As the data gram travels throughout the internet any error in header part of datagram create problems. If the destination host encounters a missing value in any field of the datagram, it discards the datagram and sends a parameter problem message back to the course. Code 0 is for when there is a error in one of header field. Code 1 is when the pointer is not used and the required part is missing.

- Query Message are used to diagnose network problems. And the two types are echo request reply and timestamp request reply.



```
Command Prompt
C:\Users\JAY>ping www.ust.hk
Pinging www.ust.hk [143.89.14.2] with 32 bytes of data:
Reply from 143.89.14.2: bytes=32 time=243ms TTL=46
Reply from 143.89.14.2: bytes=32 time=247ms TTL=46
Reply from 143.89.14.2: bytes=32 time=241ms TTL=46
Reply from 143.89.14.2: bytes=32 time=262ms TTL=46
Reply from 143.89.14.2: bytes=32 time=1922ms TTL=46
Reply from 143.89.14.2: bytes=32 time=272ms TTL=46
Request timed out.
Reply from 143.89.14.2: bytes=32 time=244ms TTL=46
Reply from 143.89.14.2: bytes=32 time=253ms TTL=46
Reply from 143.89.14.2: bytes=32 time=240ms TTL=46

Ping statistics for 143.89.14.2:
    Packets: Sent = 10, Received = 9, Lost = 1 (10% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 240ms, Maximum = 1922ms, Average = 438ms

C:\Users\JAY>ipconfig/all
'ip' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\JAY>ipconfig/all

Windows IP Configuration

Host Name . . . . . : HP
Primary Dns Suffix . . . . . : 
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : tx.rr.com

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Realtek PCIe #8 family controller
Physical Address. . . . . : 08-00-3C-07-E7-66
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 3:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : C2-3E-96-7E-01-45
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : tx.rr.com
Description . . . . . : Realtek RTL8723BE 802.11 b/g/n Wi-Fi Adapter
Physical Address. . . . . : C8-3E-96-7E-01-45
DHCP Enabled. . . . . : Yes
```